Rank 3 over a quadratic field

Problèmes ouverts — A. Dujella · 2026

Problem 2.8 (Dujella, 2026). Is there an elliptic curve over a quadratic field with torsion group $\mathbb{Z}/13\mathbb{Z}$, $\mathbb{Z}/15\mathbb{Z}$ or $\mathbb{Z}/18\mathbb{Z}$ and rank $\geqslant 3$?

Remark. For all 26 admissible torsion groups for an elliptic curve over a quadratic field, there is a curve with that torsion and rank $\geqslant 2$. For all of them, except possibly $\mathbb{Z}/13\mathbb{Z}$, $\mathbb{Z}/15\mathbb{Z}$ and $\mathbb{Z}/18\mathbb{Z}$, there is a curve with rank $\geqslant 3$ (see `https://web.math.pmf.unizg.hr/~duje/tors/torsquad.html`).